RESIDUALS, r^2 & MODEL FIT

new 5.4–5.5 • old 2.7–2.8 • ★ old 2.9, 9.1–9.5

ZONE A · THE FORMULAS + THE PLOTS

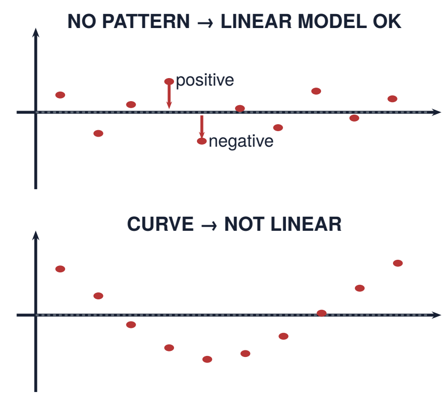
$$\text{residual} = y - \hat{y} = \text{actual} - \text{predicted}$$

r^2 = fraction of the variation in y explained by the linear relationship with x

$$s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n-2}}$$

s is the **typical size of a residual**. It has the same units as the response variable.

READ THE RESIDUAL PLOT



ZONE B · COMPUTER-OUTPUT DECODER

Predictor	Coef	SE Coef	T	P	Read it as
Constant	12.41	1.86	6.67	<.001	$\longrightarrow a$, the intercept
Study hours	4.83	0.72	6.71	<.001	$\longrightarrow b,\ s_b,\ t,\ P$-value

$S = 3.17$ $\longrightarrow s$ R-Sq $= 78.4\%$ $\longrightarrow r^2$

ZONE C · SAY IT IN WORDS

"About _____% of the variation in __________ is explained by the linear relationship with _________."

"The actual _________ was _____ units ______ what the line predicted."

ZONE D · WATCH OUT + ● BEYOND

MODEL-FIT RULES

No extrapolation: do not predict beyond the observed x-range.

r^2 gives the explained fraction, not the direction. Use r for direction.

★ BEYOND THE EXAM

Departures: outlier · high leverage · influential point

$\log(\hat{y}) = a + bx \ \Rightarrow\ \hat{y} = 10^{a+bx}$

Slope inference $(df = n-2)$:

$\mu_b = \beta \qquad \sigma_b = \dfrac{\sigma}{\sigma_x\sqrt{n}}$

$s_b = \dfrac{s}{s_x\sqrt{n-1}} \qquad t = \dfrac{b}{s_b} \qquad b \pm t^* s_b$

see WS old u2_l7–9 · u9_l1–5